REVIEW ARTICLE

Rosacea is associated with Helicobacter pylori: a systematic review and meta-analysis

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Abstract

Background Rosacea is a common skin disease characterized by facial erythema, telangiectasia, papules and pustules. Helicobacter pylori infection has been suggested to play a role in the etiopathogenesis of rosacea.

Objective To systematically review and meta-analyse the relationship between rosacea and infection with Helicobacter pylori.

Methods A literature search was performed using PubMed, EMBASE and Web of Science. Data extraction and analyses were performed on descriptive data. Study quality was assessed using the Newcastle–Ottawa Scale. Random-effects models with DerSimonian–Laird methods were utilized to estimate pooled odds ratios (ORs), with 95% confidence intervals (95% CIs). Heterogeneity of results was assessed using I^2 statistics.

Results A total of 454 articles were identified and 42 full-text articles were chosen for further review. Fourteen studies were included in the quantitative meta-analysis, comprising a total of 928 rosacea patients and 1527 controls. The overall association between Helicobacter pylori infection and rosacea was non-significant (OR 1.68, 95% CI 1.00–2.84, $P = 0.052$), but analysis restricted to *C-urea breath test* showed a significant association (OR 3.12, 95% CI 1.92–5.07, $P < 0.0001$). Effect of eradication treatment on rosacea symptoms was assessed in seven studies, but without significant effect (RR 1.28, 95% CI 0.98–1.67, $P = 0.069$).

Conclusion This meta-analysis found weak associations between rosacea and Helicobacter pylori infection as well as an effect of Helicobacter pylori therapy on rosacea symptoms, albeit that these did not reach statistical significance. Whether a pathogenic link between the two conditions exists, or whether Helicobacter pylori infection represents a proxy for other factors remains unknown.

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Declaration of interests

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Introduction

Rosacea is a common, sometimes hereditary, chronic inflammatory facial dermatosis that predominantly affects adult women[1]. It is clinically characterized by transient or persistent facial erythema, telangiectasia, papules, pustules and/or phymatous changes[1]. The estimated prevalence ranges from 1 to 20%, but rosacea is much less common in dark-skinned individuals[1]. Aggravating factors include consumption of alcohol, caffeine, smoking and ultraviolet exposure[1].

Although the underlying aetiology of rosacea is not yet fully understood, several studies have suggested a relationship with Helicobacter pylori infection and small intestinal bacterial overgrowth (SIBO)[2–7].

Helicobacter pylori, a spiral shaped, Gram-negative, microaerophilic bacteria, is predominantly found in human gastric mucosa and has been linked to gastrointestinal (GI), dermatological and neurological diseases.[8–10] In textbooks, experts have had different opinions about the relationship of Helicobacter

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pylori and rosacea that have varied between coincidental, plausible or undecided[11–13].

The prevalence of Helicobacter pylori varies between study population and applied diagnostic methods. As much as one-third of Northern European and Northern American populations, and up to 50% of people in Southern Europe, South America and Asia are colonized with Helicobacter pylori[14]. Importantly, previous studies have compared the prevalence of Helicobacter pylori infection in rosacea patients and controls and also investigated effects of Helicobacter pylori treatment on rosacea symptoms and severity[3,4,15–19]. In a recent study by Saleh *et al.*,[20] 138 (92%) rosacea patients with a positive Helicobacter pylori stool antigen who underwent Helicobacter pylori eradication treatment had a negative stool antigen at the end of the trial. As the relationship between rosacea and Helicobacter pylori remains unresolved, we performed a systematic review and meta-analysis to examine whether the two conditions are associated.

Methods

Selection of studies

Two authors (A-H.R.J. and R.G.) systematically searched the PubMed, EMBASE and Web of Science databases using the following search terms: PubMed (Rosacea[Mesh] OR Rosacea) AND (Helicobacter[mesh] OR helicobacter infections[mesh] OR helicobacter*), Embase (rosacea.af AND helicobact*.af), Web of Science (ts=rosacea AND ts=helicobact*). Our search was limited to English-language and human-only studies, published between 1 January 1990 and 1 January 2016. The search yielded 252 non-duplicate results. Titles/abstracts were read to determine eligibility for inclusion in the systematic review. To be included, original studies fulfilled the following inclusion criteria: case–control, cross-sectional, cohort or nested case–control design; evaluation of Helicobacter pylori infection in conjunction with rosacea; and analyses that compared rosacea patients with control groups.

Data extraction

The Meta-Analysis of Observational Studies in Epidemiology (MOOSE) guidelines were used to guide the analysis[21]. The systematic review and data extraction were performed independently by two screeners (A-H.R.J. and R.G.), and any differences were adjudicated by A.E. and J.P.T. For each study included, we recorded the study year, country in which the study population lived, setting in which the study took place, study design, numbers of case and control subjects, age, sex and data collection processes including method for identifying the presence Helicobacter pylori. The Newcastle–Ottawa Scale[22] was used to assess study quality by categorizing into three dimensions: selection, comparability and exposure for case–control studies. The selection, comparability and exposure dimensions contain four, two and three items, respectively. A star system was used as a semi-quantitative assessment of study quality. A study was awarded a maximum of one star for each numbered item within the selection and exposure categories. A maximum of two stars were awarded for comparability. Studies with a score of 0–5 were categorized as lower quality, whereas studies with scores of 6 or above were categorized as higher quality. The primary outcome was the association of rosacea with Helicobacter pylori infection for each study. As a secondary outcome, the effect of treatment of Helicobacter pylori infection was assessed in some of the included studies. Data extraction was performed by A-H.R.J. For each of the studies, the type of medication and duration of its use were recorded, as well as previous treatments. Finally, the effect of treatment was registered.

Statistical analysis

We assessed between-study heterogeneity using I^2 tests, where substantial heterogeneity was considered with I^2 values >50%. We used random-effects (DerSimonian and Laird) models to compute the pooled odds ratio (OR) of association between rosacea and Helicobacter pylori infection. Funnel plots were visually inspected to asses for publication bias. All analyses were performed using StatsDirect version 3.0.190 (StatsDirect Ltd., Cheshire, UK). P-values of <0.05 were considered significant, and results were reported with 95% confidence intervals (95% CIs).

Results

Study selection

From the initial 454 search results, 42 articles were chosen for full-text review. Among these, 28 studies were excluded, because they were review articles, duplicates or single case reports (Fig. 1). Fourteen studies were included in the final analysis, including a total of 928 rosacea patients and 1527 controls. All studies were published between 1998 and 2015; six were from Southern Europe[23–28], two from Northern Europe[4,29], two from the United States[19,30], and one study from Egypt[31], Nepal[32], Korea[33] and Brazil[2], respectively. Seven of the included studies also examined the effect of therapy on rosacea-related skin symptoms. Four of these studies were from Southern Europe[23,24,27,28], one study was from Northern Europe[4], one from Egypt[31] and one from Korea[33]. The study characteristics are shown in Table 1.

Quality of studies and publication bias

Newcastle–Ottawa Scale (NOS) was used to assess study quality. Of the 14 studies included, only one study did not meet a score of six or more. Scores from six to nine were awarded to the remaining 13 studies.

The analysis of pooled studies showed a significant heterogeneity (I^2 = *85.5%*), but visual assessment of the funnel plot did not suggest significant publication bias (Fig. 2).

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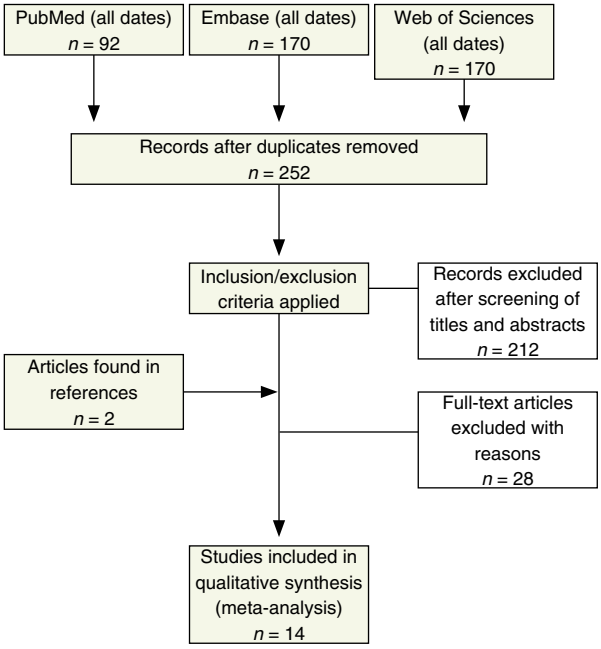

Figure 1 Study flow chart.

Association between Helicobacter pylori infection and rosacea

Helicobacter pylori infection was found in 520 (56%) of 928 of rosacea patients and 748 (49%) of 1527 controls. The combined OR of Helicobacter pylori infection in rosacea patients compared to controls was 1.68, 95% CI 1.00–2.84, $P = 0.052$ (Fig. 3). Subgroup analysis revealed a clear trend towards a stronger association between Helicobacter pylori and rosacea in studies using *C-urea breath test* (C-UBT) to establish Helicobacter pylori infection (fixed effects OR 3.12, 95% CI 1.92–5.07, $P < 0.0001$). A positive association between rosacea and Helicobacter pylori was found in studies from all regions that were studied, but it was non-significant in any region when analysed separately (Northern Europe: OR 1.37, 95% CI 0.18–10.68; Southern Europe: OR 1.51, 95% CI 0.54–4.22; other study locations: OR 1.72, 95% CI 0.81–3.67). The effect of treatment of Helicobacter pylori infection on skin symptoms was assessed in seven studies. A non-significant tendency towards improvement of rosacea-related skin symptoms was observed in patients who underwent such therapy (RR 1.28, 95% CI 0.98–1.67, $P = 0.0685$) (Fig. 4).

Discussion

Main findings

In this systematic review and meta-analysis, we found a non-significant association between rosacea and Helicobacter pylori infection. Studies using the C-UBT for the diagnosis of Helicobacter pylori infection, which is known to have greater diagnostic value compared to serology[8,18,34], found a stronger association between the two conditions compared with studies using serological tests. As a secondary finding, a non-significant improvement of skin symptoms in rosacea was observed after treatment of Helicobacter pylori.

Interpretation

Improved hygiene and increased use of antibiotics, as well as changing demographics with a tendency towards smaller families, represent explanatory factors for the reduced prevalence of Helicobacter pylori infection over the past decades in the industrialized parts of the world[35]. Notably, Helicobacter pylori urease activity, and the subsequent rise in gastric pH, may result in a

Table 1 Characteristics of studies included in meta-analysis

Author	Year	Country	Cases	Positive/ cases	Controls	Positive/ controls	Serology (IgG)	Breath test	Biopsy, histology	Stool sample
Colgecen	2015	Turkey	21	16	30	14		x		
Gravina	2014	Italy	90	44	90	24		x		x
El-Khalawany	2012	Egypt	68	49	54	25			x	
Bhattarai	2012	Nepal	26	17	52	6	x			
Abram	2009	Estonia	170	106	217	113	x			
Baz	2004	Turkey	29	3	20	13	x			
Gurer	2002	Turkey	33	29	20	9	x			
Szlachcic	2002	Poland	60	53	60	39	x	x	x	
Herr	2000	Korea	50	42	50	39			x	
Bonamigo	2000	Brazil	62	41	124	72	x			
Rojo-Garcia	2000	Spain	197	79	547	307	x			
Utas	1999	Turkey	25	17	87	42	x		x	
Sharma	1998	USA	45	12	43	15	x			
Jones	1998	USA	52	12	133	30	x (cases)		x (controls)	

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Table 2 Newcastle–Ottawa Scale (NOS) Quality Assessment Table

Author	Selection	Comparability	Exposure	Total NOS score
Colgecen (2015)	***	**	***	8
Gravina (2014)	***	**	***	8
El-Khalawany (2012)	*	*	***	5
Bhattarai (2012)	***	**	***	8
Abram (2009)	****	**	***	9
Baz (2004)	**	*	***	6
Gurer (2002)	**	*	***	6
Szlachcic (2002)	**	**	***	7
Herr (2000)	**	**	***	7
Bonamigo (2000)	**	*	***	6
Rojo-Garcia (2000)	***	*	***	7
Utas (1999)	***	*	***	7
Sharma (1998)	****	*	***	8
Jones (1998)	***	*	**	6

A star system was used for allow a semiquantitative assessment of study quality. A study was awarded a maximum of one star for each numbered item within the selection and exposure categories. A maximum of two stars were awarded for comparability. The NOS ranges from zero to nine stars. We considered high-quality studies as those that achieved six or more stars and poor-quality studies those with fewer than six stars.

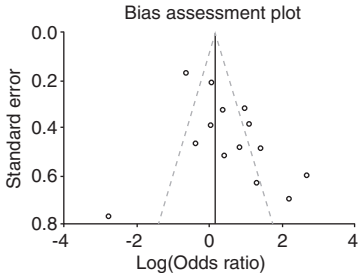

Figure 2 Bias assessment plot.

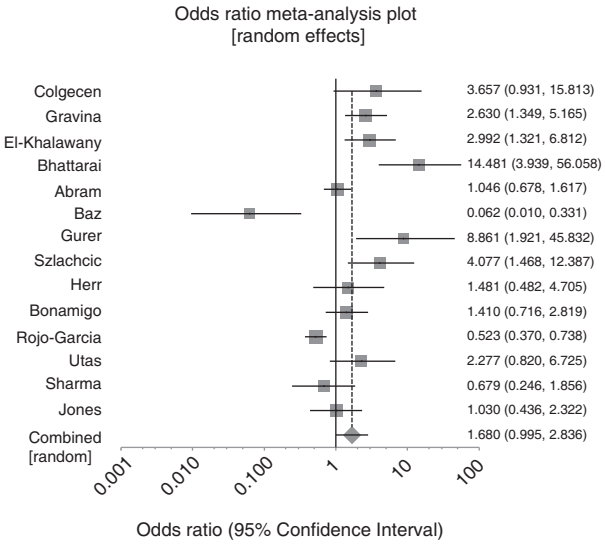

Figure 3 Meta-analysis of the risk of Helicobacter pylori infection in patients with rosacea compared to controls.

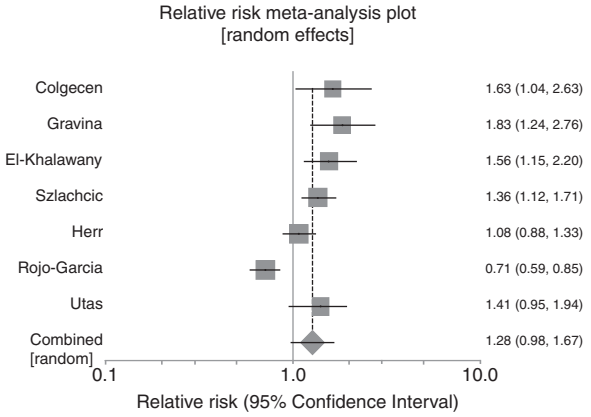

Figure 4 Meta-analysis of the effect of Helicobacter pylori eradication on rosacea symptoms.

more favourable environment for certain but not all bacteria, and including some that may worsen rosacea symptoms[36]. Therefore, it is possible that Helicobacter pylori infection reflects the changing humane microecology and could represent a proxy for hitherto unidentified pathogenic factors.

In a pioneering study including 60 patients with rosacea, Watson *et al.*[37] found an association between rosacea and GI diseases. A total of 20 patients had abnormal jejunal mucosa, four had coeliac disease (CeD), 34 patients suffered from dyspepsia, and 16 had a history of diarrhoea[37]. In a case–control study including 65 patients with rosacea and 65 controls[38], a significant association between severity of rosacea and gastroesophageal reflux disease and other GI diseases was observed. In a recent nationwide study from Denmark, rosacea was associated with an increased prevalence of CeD, Crohn's disease (CD), ulcerative colitis (UC), Helicobacter pylori infection, small intestinal bacterial overgrowth (SIBO) and irritable bowel syndrome (IBS)[39]. Moreover, a large US observational case–control study reported an increased risk of developing rosacea in patients with inflammatory bowel disease (IBD), particularly among those with UC[40]. Indeed, shared genetic risk loci for rosacea and CeD have been identified suggestive of a pathogenic links[41]. SIBO is defined as unexpected microbial concentration (>10^5 colony-forming units/mL) in jejunal aspirate culture[7]. In a study by Parodi *et al.*,[42] a significantly higher prevalence of SIBO was found in rosacea patients compared to controls. In the same

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study, eradication of SIBO was associated with a significant improvement of rosacea skin symptoms. However, in a study by Gravina *et al.*,[43]no difference in SIBO prevalence was observed between rosacea patients and controls. Collectively, these data suggest that there is a positive relationship between rosacea and GI diseases and that Helicobacter pylori infection could be one of several causative factors, or a proxy for associated lifestyle factors, presence of other pathogenic bacteria or underlying inflammation, respectively.

Importantly, emerging epidemiological evidence suggests that rosacea may be associated with neurological diseases, including Parkinson's disease[44,45] and Alzheimer's dementia[46]. In an older German case–control study, including 70 patients with Parkinson's disease and 22 controls, rosacea was found in 13 (18.6%) patients[47]. While the underlying cause remains to be elucidated, there could be a link between the gut, skin and brain. Indeed, in patients with Parkinson's disease, the entire GI tract appears to be affected by this disease and the gut may be where the disease originates[48]. Additionally, persistent SIBO and Helicobacter pylori colonization have also been suggested to contribute in the development of GI dysfunction in Parkinson's disease[49]. Nevertheless, while these hypotheses are intriguing, further research is required to determine whether a place for rosacea exists in the gut–skin–brain axis.

Strengths and limitations

Data from several independent study populations and from different parts of the world were included in this meta-analysis, making the results relatively generalizable. Both presence of Helicobacter pylori infection and validation of attempted eradication therapy were examined and various diagnostic methods evaluated. Still, this meta-analysis had important limitations. Overall, a low number of patients and studies were included. Moreover, several studies did not mention whether the patients or controls had previously received treatment for Helicobacter pylori infection, potentially resulting in falsely low Helicobacter pylori infection rate, and lastly, no information on previous antibiotic therapy for rosacea was given. One weakness of this study is that it could not analyse different strains of Helicobacter pylori some of which are more virulent than others and that the effect of eradication therapy was not examined. In the study by Szlachcic[50], it was determined that Helicobacter pylori with CAG-A toxin was associated with rosacea. This form is more virulent in general and is more associated with gastric cancer than other Helicobacter pylori.

Conclusion

In conclusion, our meta-analysis found evidence of a weak, but non-significant, association between rosacea and Helicobacter pylori infection. As a secondary finding, a non-significant tendency towards improvement of rosacea-related skin symptoms was observed following Helicobacter pylori therapy. It remains unknown whether Helicobacter pylori infection could represent a proxy for growth of other intestinal bacteria or underlying immune alterations. Future studies should therefore examine other factors than Helicobacter pylori colonization to control for possible confounding, including the presence of subtypes of Helicobacter pylori.

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